

[illegible]

ESP32-WROVER-IE MODULE

ESPRESSIF

U8

Power and Ground Connections:

- +3.3V to Pin 1 (VDD33(3V3))
- +3.3V to Pin 39 (THERMAL_PAD)
- GND to Pin 2 (GND)
- GND to Pin 4 (GND)
- GND to Pin 5 (GND)
- GND to Pin 38 (GND)
- GND to Pin 37 (GND)
- GND to Pin 36 (GND)
- GND to Pin 35 (GND)
- GND to Pin 34 (GND)
- GND to Pin 33 (GND)
- GND to Pin 32 (GND)
- GND to Pin 31 (GND)
- GND to Pin 30 (GND)
- GND to Pin 29 (GND)
- GND to Pin 28 (GND)
- GND to Pin 27 (GND)
- GND to Pin 26 (GND)
- GND to Pin 25 (GND)
- GND to Pin 24 (GND)
- GND to Pin 23 (GND)
- GND to Pin 22 (GND)
- GND to Pin 21 (GND)
- GND to Pin 20 (GND)
- GND to Pin 19 (GND)
- GND to Pin 18 (GND)
- GND to Pin 17 (GND)
- GND to Pin 16 (GND)
- GND to Pin 15 (GND)
- GND to Pin 14 (GND)
- GND to Pin 13 (GND)
- GND to Pin 12 (GND)
- GND to Pin 11 (GND)
- GND to Pin 10 (GND)
- GND to Pin 9 (GND)
- GND to Pin 8 (GND)
- GND to Pin 7 (GND)
- GND to Pin 6 (GND)
- GND to Pin 5 (GND)
- GND to Pin 4 (GND)
- GND to Pin 3 (GND)
- GND to Pin 2 (GND)
- GND to Pin 1 (GND)

Signal Connections:

- Pin 10 (GPIO12/ADC2_CH5/TOUCH5/RTC_GPIO15/MTDI/HSP10/H52_DATA2/SD_DATA2/EMAC_TX_CLK) to Pin 10 (GPIO12/ADC2_CH5/TOUCH5/RTC_GPIO15/MTDI/HSP10/H52_DATA2/SD_DATA2/EMAC_TX_CLK)
- Pin 11 (GPIO13/ADC2_CH6/TOUCH6/RTC_GPIO16/MTMS/HSP1CLK/H52_CLK/SD_CLK/EMAC_TX_CLK) to Pin 11 (GPIO13/ADC2_CH6/TOUCH6/RTC_GPIO16/MTMS/HSP1CLK/H52_CLK/SD_CLK/EMAC_TX_CLK)
- Pin 12 (GPIO14/ADC2_CH7/TOUCH7/RTC_GPIO17/H51_STROBE) to Pin 12 (GPIO14/ADC2_CH7/TOUCH7/RTC_GPIO17/H51_STROBE)
- Pin 13 (GPIO15/ADC2_CH3/TOUCH3/RTC_GPIO18/HSP1CS0/SD_CMD/EMAC_RX_CLK) to Pin 13 (GPIO15/ADC2_CH3/TOUCH3/RTC_GPIO18/HSP1CS0/SD_CMD/EMAC_RX_CLK)
- Pin 14 (GPIO16/ADC2_CH4/TOUCH4/RTC_GPIO19/H51_DATA4/SD_DATA4/EMAC_RX_CLK) to Pin 14 (GPIO16/ADC2_CH4/TOUCH4/RTC_GPIO19/H51_DATA4/SD_DATA4/EMAC_RX_CLK)
- Pin 15 (GPIO17/H51_DATA5/SD_DATA5/EMAC_RX_CLK) to Pin 15 (GPIO17/H51_DATA5/SD_DATA5/EMAC_RX_CLK)
- Pin 16 (GPIO18/HSP1CLK/H51_DATA7) to Pin 16 (GPIO18/HSP1CLK/H51_DATA7)
- Pin 17 (GPIO19/VSQIP/U0CTS/EMAC_TX0) to Pin 17 (GPIO19/VSQIP/U0CTS/EMAC_TX0)
- Pin 18 (GPIO21/VSQIP/H51_STROBE) to Pin 18 (GPIO21/VSQIP/H51_STROBE)
- Pin 19 (GPIO22/VSQIP/U0TS/EMAC_TX0) to Pin 19 (GPIO22/VSQIP/U0TS/EMAC_TX0)
- Pin 20 (GPIO23/VSQIP/H51_STROBE) to Pin 20 (GPIO23/VSQIP/H51_STROBE)
- Pin 21 (GPIO25/DAC_1/ADC2_CH8/RTC_GPIO6/EMAC_RX0) to Pin 21 (GPIO25/DAC_1/ADC2_CH8/RTC_GPIO6/EMAC_RX0)
- Pin 22 (GPIO26/DAC_2/ADC2_CH9/RTC_GPIO7/EMAC_RX0) to Pin 22 (GPIO26/DAC_2/ADC2_CH9/RTC_GPIO7/EMAC_RX0)
- Pin 23 (GPIO27/ADC2_CH7/TOUCH7/RTC_GPIO17/EMAC_RX0) to Pin 23 (GPIO27/ADC2_CH7/TOUCH7/RTC_GPIO17/EMAC_RX0)
- Pin 24 (GPIO32/XTAL_32K_P/ADCL_CH4/TOUCH9/RTC_GPIO9) to Pin 24 (GPIO32/XTAL_32K_P/ADCL_CH4/TOUCH9/RTC_GPIO9)
- Pin 25 (GPIO33/XTAL_32K_N/ADCL_CH5/TOUCH8/RTC_GPIO8) to Pin 25 (GPIO33/XTAL_32K_N/ADCL_CH5/TOUCH8/RTC_GPIO8)
- Pin 26 (GPIO34/ADCL_CH6/RTC_GPIO4) to Pin 26 (GPIO34/ADCL_CH6/RTC_GPIO4)
- Pin 27 (GPIO35/ADCL_CH7/RTC_GPIO5) to Pin 27 (GPIO35/ADCL_CH7/RTC_GPIO5)
- Pin 28 (GPIO36/SENSOR_VP/ADCL_CH0/RTC_GPIO10) to Pin 28 (GPIO36/SENSOR_VP/ADCL_CH0/RTC_GPIO10)
- Pin 29 (GPIO39/SENSOR_VN/ADCL_CH3/RTC_GPIO3) to Pin 29 (GPIO39/SENSOR_VN/ADCL_CH3/RTC_GPIO3)

ESP32-WROVER & ESP32-WROOM-32 MODULES

Note:

- When ESP32-WROVER, take in mind that
 - GPIO16<27> and GPIO17<28> are not connected
- GPIO6 to GPIO11 are connected to the SPI flash integrated on the module and are not led out

ESP32-WROVER-IE

Access Bus

The diagram illustrates the Access Bus connections for the HN1x8 module. It shows a set of 8 pins on the right, labeled 1 through 8, with a label 'Access_Bus1' above them. On the left, there are four signal lines and three power lines. The signal lines are labeled: 'ESP_EN', 'GPIO2_U1TXD', 'GPIO34_U1RXD', and 'GPIO0_U1RTS'. The power lines are labeled: '+5V', '+3.3V', and 'GND'. The connections are as follows: 'ESP_EN' connects to pin 1, 'GPIO2_U1TXD' to pin 2, 'GPIO34_U1RXD' to pin 3, 'GPIO0_U1RTS' to pin 4, '+5V' to pin 5, '+3.3V' to pin 6, and 'GND' to pin 7. Pin 8 is shown but has no connection.

Signal / Power	Pin
ESP_EN	1
GPIO2_U1TXD	2
GPIO34_U1RXD	3
GPIO0_U1RTS	4
+5V	5
+3.3V	6
GND	7
	8

HN1x8

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LCD Connector

The diagram illustrates the pinout for an LCD connector. It consists of 12 pins, numbered 1 through 12. The functions and power levels for each pin are as follows:

Pin Number	Function	Power Level
1	GPI05\Blue_1\VSPL_CS	+3.3V
2	GPI022\Red_1\LCD_DC	GND
3	GPI018\Green_0\VSPL_SCK	
4	GPI023\HSync\VSPL_MOSI	
5	GPI019\Green_1\VSPL_MISO	
6	GPI021\Red_0\LCD_RST	
7	GPI015\VSync\I2C_SCL	
8	GPI04\Blue_0\I2C_SDA	
9		
10		
11	GND	
12	+5V	

The connector is labeled **LCD_CON1** at the top right and **HN1x12** at the bottom right.

[illegible]

PS/2 Port 0 -> Keyboard

PS2_KEYBOARD1

IO	1	1
NA	2	2
GN	3	3
5V	4	4
CK	5	5
NC1	6	6
NC2	7	7
SHIELD1	8	8
MDR6_MINI-DIN	9	9

Current Consumption Approximately: 50mA

+5V

R19 10k/R0402

R28 120R/R0402

GPIO32_KBD_DAT

+5V

R20 10k/R0402

R29 120R/R0402

GPIO33_KBD_CLK

C13 100nF/50V/X7R/0603

GND

PS/2 Port 1 -> Mouse

Current Consumption
Approximately: 100mA

PS2_MOUSE1

IO	1	5	IO
NA	2	4	NA
GN	3	5	GN
5V	4	5	5V
CK	5	5	CK
NA	6	5	NA
NC1	7	5	NC1
NC2	8	5	NC2
SHIELD	9	5	SHIELD

MDR6_MINI-DIN

GND GND

+5V

R34
10k/R0402

R38
120k/R0402

GPIO27_MOUSE_DAT

+5V

R35
10k/R0402

R39
120k/R0402

GPIO26_MOUSE_CLK

C18
100nF

Mounting Holes



MH1

Mounting_Hole_NPTH



MH2

Mounting_Hole_NPTH



MH3

Mounting_Hole_NPTH



MH4

Mounting_Hole_NPTH

U1

PA1/OSC1/PA1_T1SCH2/OPN0
PA2/OSC0/PA2_T1CH2N/OPPO/AETR2

PC0/T2CH3/UTX_T1SS_T1CH3
PC1/SDA/SS/T2CH4_T1CH1ETR_T1BK1N_URX
PC2/SCL/URTS_T1BK1N/AETR_T2CH2_T1ETR
PC3/T1CH3_T1CH1N_UTC5
PC4/A2/T1CH4/MC0/T1CH1CH2N
PC5/SCK/T1ETR_T2CH1ETR_USCK_UTC1/T1CH3
PC6/MOSI/T1CH1CH3N_UTC5_VSDA
PC7/MISO/T1CH2_T2CH2/URTS

■ FT = 5V tolerant

PD0/T1CH1N/OPN1/SDA_UTC1
PD1/SMIO/AETR2/T1CH3N/SCL_URX
PD2/A3/T1CH1/T2CH3_T1CH2N
PD3/A4/T2CH2/AETR/UTC5/T1CH4
PD4/A7/UTC/T2CH1ETR/OPD_T1CH1ETR
PD5/A5/UTX/T2CH4_URX
PD6/A6/URX/T2CH3_UTC
PD7/NRST/T2CH4/OPP1/UCK

PA15/SoftSPI_MOSI
PA2/SoftSPI_MISO

SPI1_NSS(PC0)
I2C1_SDA(PC1)
I2C1_SCL(PC2)
PC3/EXT_PWR_EN#

SPI1_SCK(PC5)
SPI1_MOSI(PC6)
SPI1_MISO(PC7)

EXT1_Line0(PD0)
PD0/PWR_SENS
PD1/SMIO
PD2/BAT_SENS
PD3/SoftSPI_CS#
PD4/SoftSPI_SCK

USART1_Tx(PD5)
USART1_Rx(PD6)
PD7_RST#

PD5/SMIO

U6

Y
VCC Power GND

5N74LVC14(0400BVR(SOT23-5)
C8

100nF/10V/10%/X7R/C0402

GPIO13_HSP1CS

GPIO13V_EXTP_IRQ
GPIO14_HSP1_CLK
GPIO12_HSP1_MOSI
GPIO13V_HSP1_MISO

PGM/DBG

NA(H1x12)

CH32V003F4P6TSSOP20

pander

UEXT_PWR_EN# to be included by default!

PC3\UEXT_PWR_EN#

R30 2.2k/0402

R31 220k/0402

C17 33pF/0402/0402

FET2 WPM2015-3/TR

+3.3V

R32 2.2k/0402

UEXT1

R37 2.2k/0402

GND

PD5\UTX
PC2\SCL
PA2\SoftSPL_MISO
PD4\SoftSPL_SCK

PD6\URX
PC3\SDA
PA1\SoftSPL_MOSI
PD3\SoftSPL_CS#

B-R-10-LF (BH10R)

Fiducials

 FID1 Fiducial	 FID3 Fiducial	 FID5 Fiducial
 FID2 Fiducial	 FID4 Fiducial	 FID6 Fiducial



OLIMEX
<https://www.olimex.com>
OLIMEX LTD.
 Sheet: /
 File: working.kicad_sch
Title: ESP32-SBC-FabGL

Rev: A
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